

WKB approximation continued

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We have the expression for the classically allowed region to be

$$\psi(x) \approx \frac{c}{\sqrt{p(x)}} e^{\pm \frac{i}{\hbar} \int p dx} \quad (1)$$

with $\psi(0) = \psi(a) = 0$ quantisation from constraint.

$$\sqrt{2m} \int_0^a dx \sqrt{E_k - V(x)} = k\hbar\pi \quad (2)$$

We can look at an example from griffiths

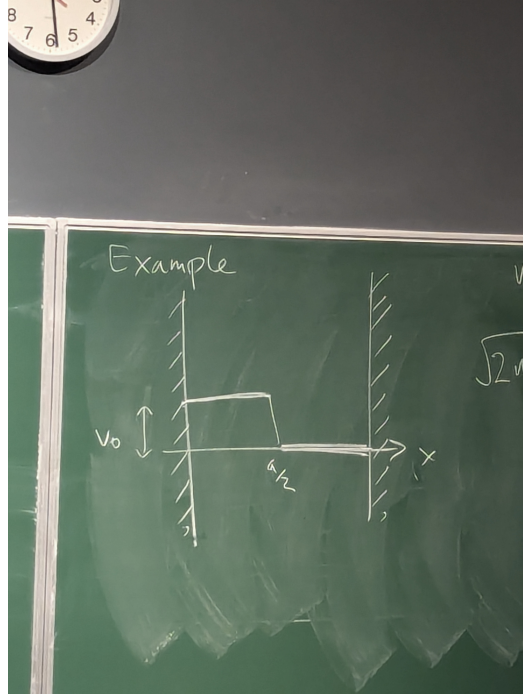


Figure 1: XXX.jpg

WKB quantisation

$$\sqrt{2m} \left(\int_0^{\frac{a}{2}} \sqrt{E_k - V_0} dx + \int_{\frac{a}{2}}^a \sqrt{E_k} dx \right) = k\hbar\pi \quad (3)$$

$$\sqrt{2m} \left(\frac{a}{2} \sqrt{E_k - V_0} + \frac{a}{2} \sqrt{E_k} \right) = k\hbar\pi \quad (4)$$

$$\sqrt{E_k - V_0} + \sqrt{E_k} = \frac{\sqrt{2}k\hbar\pi}{a\sqrt{m}} \quad (5)$$

We call the rhs γ_k

$$\sqrt{E_k - V_0} + \sqrt{E_k} = \gamma_k \quad (7)$$

$$E_k - V_0 - 2\sqrt{E_k}\sqrt{E_k - V_0} + E_k = \gamma_k^2 \quad (8)$$

$$2E_k - V_0 + 2\sqrt{E_k}(\gamma_k - \sqrt{E_k}) = \gamma_k^2 \quad (9)$$

$$2\sqrt{E_k}\gamma_k - V_0 = \gamma_k^2 \quad (10)$$

$$\sqrt{E_k} = \frac{\gamma_k^2 + V_0}{2\gamma} \quad (11)$$

$$E_k = \frac{(\gamma_k^2 + V_0)^2}{4\gamma_k^2} \quad (12)$$

Now if we consider the classically forbidden region

$$E < V(x), \psi^2(x) < 0 \quad (13)$$

$$|p(x)| = \sqrt{2m(V(x) - E)} \quad (14)$$

repeat same steps as for the previous case and we obtain

$$\psi(x) = \frac{c}{\sqrt{|p(x)|}} e^{\pm \frac{i}{\hbar} \int |p(x)| dx} \quad (15)$$

We pay attention the sign. to ensure that we have a normalization-able function we need to pick the exponential with a negative power.

The classically forbidden sol is

$$\psi(x) = \frac{c}{\sqrt{|p(x)|}} e^{-\frac{i}{\hbar} \int |p(x)| dx} \quad (16)$$

Things get messy at boundary of classically allowed and not allowed regions. Complicated things with airy functions typically done on mathematica.